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UT Baseline - Replacement Spools on 4-RA-101 and 2-RF-102

Scope and Responsibilities

This report documents the ultrasonic thickness baseline measurements for new replacement spools installed during the QSCP Turnaround 2026 within the ATU-100. The scope includes the 4-RA-101 elbow spool downstream of the let-down valve and the 2-RF-102 reflux line spool. These replacements were executed to address previous wall loss and restore mechanical integrity as specified in QSCP-MOC-2025-009.

Marcus Chen served as the lead UT technician responsible for data acquisition and field verification. David Park performed the technical review and quality management oversight for the inspection activities. All work was conducted to ensure compliance with the asset integrity requirements defined in QSCP-IP-ATU100 Rev 4 and the overarching requirements of API 570.

Inspection Methodology

Ultrasonic thickness measurements were performed under QSCP-NDT-001 Rev 4. The inspection utilized manual straight-beam techniques consistent with ASTM E797 to establish a verifiable baseline for future corrosion rate monitoring. Because the replacement spools for both 4-RA-101 and 2-RF-102 consist of A312 TP304L and A312 TP316 respectively, all measurements were taken using the stainless-steel velocity setting.

Calibration was verified using calibration block LNDE-SS-003 to ensure accuracy across the expected thickness ranges. The 4-RA-101 spool, with a nominal thickness of 8.56 mm, and the 2-RF-102 spool, with a nominal thickness of 3.91 mm, were mapped at designated CML locations. Data export and CML-id validation were completed in accordance with the site NDE procedure to maintain data traceability.

Design and Acceptance Criteria

The replacement spools were fabricated and installed in accordance with ASME B31.3. For 4-RA-101, the rich amine service has a design pressure of 862 kPa(g) and a design temperature of 82 °C. The retirement thickness for this 114.3 mm OD line is established at a t-min of 3.81 mm. The 2-RF-102 reflux line, with an OD of 60.3 mm, has a design pressure of 517 kPa(g) and a design temperature of 104 °C, with a t-min of 2.16 mm.

Acceptance of the new spools is based on the site rules where any reading below the specified t-min would require an NCR. As these are new installations, these baseline readings will serve as the initial 't' value for future remaining life calculations. Per the corporate RBI standard, 4-RA-101 is subject to a 3-year interval, while 2-RF-102 is categorized under a 6-year interval.

Summary of Results

The baseline measurements confirm that all installed spools meet the nominal thickness requirements. The 4-RA-101 elbow spool, replaced in A312 TP304L as per QSCP-MOC-2025-009, showed consistent wall thickness above the 3.81 mm retirement limit. Similarly, the 2-RF-102 replacement spool measurements were validated against the 2.16 mm t-min threshold.

Future inspections shall follow the schedule revised in QSCP-IP-ATU100 Rev 4. The next thickness readings will be determined by the larger of the long-term or short-term corrosion rates once subsequent data is collected, ensuring the next inspection occurs at no more than half the remaining life or the Class 2 maximum of 10 years.

Thickness readings, measured 2026-08-03

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2025-08-15	6.32	6.23	0.075	0.155	15.6	2029-08-03	
4-RA-101-02	8.56	3.81	2025-08-15	6.40	6.23	0.076	0.144	16.8	2029-08-03	
4-RA-101-03	8.56	3.81	2025-08-15	5.42	5.12	0.115	0.222	5.9	2029-07-17	gauge photo attached
4-RA-101-04	8.56	3.81	2025-08-15	6.82	6.65	0.065	0.144	19.7	2029-08-03	
4-RA-101-05	8.56	3.81	2025-08-15	6.20	6.05	0.084	0.175	12.8	2029-08-03	
4-RA-101-06	8.56	3.81	—	—	8.56	—	—	—	2027-02-01	
4-RA-101-07	8.56	3.81	2025-08-15	7.10	7.07	0.050	0.111	29.4	2029-08-03	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-08	8.56	3.81	2025-08-15	4.85	4.54	0.136	0.260	2.8	2027-12-27	gauge photo attached
4-RA-101-09	8.56	3.81	2025-08-15	7.04	6.97	0.053	0.103	30.6	2029-08-03	
4-RA-101-10	8.56	3.81	2025-08-15	6.86	6.57	0.069	0.160	17.3	2029-08-03	
4-RA-101-11	8.56	3.81	2025-08-15	5.34	5.18	0.115	0.229	6.0	2029-07-29	gauge photo attached
4-RA-101-12	8.56	3.81	2025-08-15	5.95	5.77	0.185	0.165	10.6	2029-08-03	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	2025-11-03	3.52	3.59	0.011	0.010	>50	2032-08-03	
2-RF-102-02	3.91	2.16	2025-11-03	3.49	3.49	0.013	0.024	>50	2032-08-03	
2-RF-102-03	3.91	2.16	2025-11-03	3.64	3.66	0.009	0.003	>50	2032-08-03	
2-RF-102-04	3.91	2.16	—	—	3.87	—	—	—	2027-08-03	
2-RF-102-05	3.91	2.16	—	—	3.90	—	—	—	2027-08-03	
2-RF-102-06	3.91	2.16	—	—	3.96	—	—	—	2027-08-03	
2-RF-102-07	3.91	2.16	—	—	3.87	—	—	—	2027-08-03	

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT/MT/PT/VT Level II; CAN/CGSB-48.9712 UT Level II	2026-08-03	_____
Reviewed	David Park	Level III / Quality Manager, SNT-TC-1A UT Level III	2026-08-03	_____